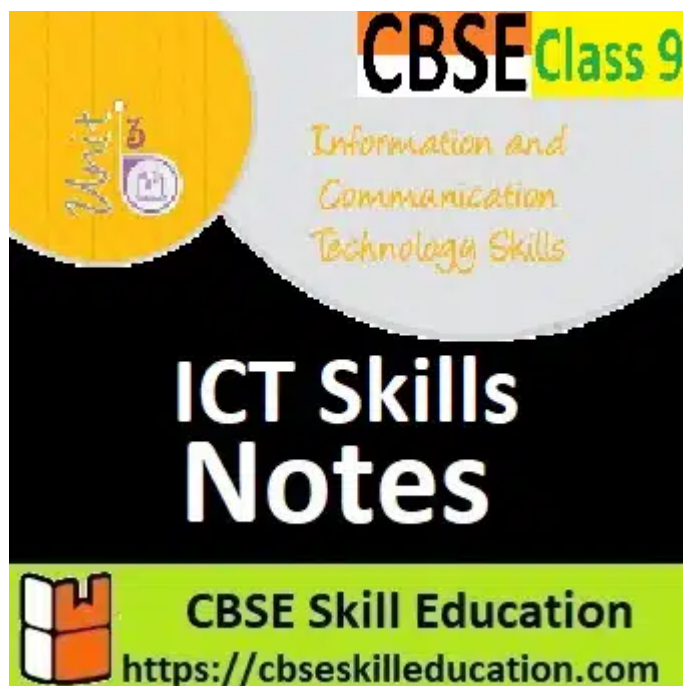


Basic ICT Skills Class 9 Notes

PART A - EMPLOYABILITY SKILLS | UNIT 03 BASIC ICT SKILLS | NOTES

Source: <https://cbse skillededucation.com/basic-ict-skills-class-9-notes/>

Basic ICT Skills Class 9 Notes are designed to help students understand the fundamentals of Information and Communication Technology (ICT) used in everyday life. In today's digital world, ICT skills are essential for education, communication, and future career opportunities.



Information and communication technology is referred to as ICT. We can communicate, manage our businesses, and stay connected with our family and friends using ICT.

Session 1: Introduction to ICT

ICT help you to effectively utilise the knowledge and skills. The basic ICT skills that you need are

- How to operate computers
- How to browse the Internet for collecting, storing the information.

Places where ICT is used

- **ICT at Workplace** – In workplace computer software and applications is used to making documents, tables, graphs and to do calculation.
- **ICT at Home** – In home people are use television for entertainment and phones for calling up other people.
- **ICT at Agriculture** – Different agriculture related activities like research, development and training is done using ICT.
- **ICT at Banking and Finance** – ICT helps banks to provide better services to customers, online banking system, customers can access the accounts etc.

Session 2: ICT Tools: Smartphones and Tablets – I

Many people carry smartphones and tablets with them and use them for everyday tasks like sharing photos and sending emails, these devices are becoming more and more important. Without these, most people cannot work in offices and home.

Smartphones

Smartphones are also known as mobile phone, these mobile phone can be used at home or in office. Smartphone do not need telephone line for communication, they use wireless mobile network to make calls and to connect to the internet.

Some of the popular **operating systems for smartphones are Android OS, Apple iOS and Windows Mobile.**

Tablets

A tablet is small computers that combines input, output, and processing capabilities into a single “touchscreen.” You may perform a variety of tasks on a tablet just by touching the screen. You may use it for all the same purposes as a computer or smartphone.

TV and Radio

TV and radio are used as an ICT tool from a long time. While TV is used to broadcast audio-visual content to a large audience, radio is used to broadcast audio.

Applications or apps

Apps are software applications that carry out various tasks. They can be defined as a collection of modules, procedures, or instructions that enable a certain kind of computer operation. The phone or tablet already has some of the apps. We refer to these as “default” apps.

Commonly found applications in Smartphones and tablet

App Name	Working properties
Phone	This app is used to make calls.
Camera	You can takes photos and videos using camera.
Mail	You can send mail or recieve mail in the mobile phone using e-mail app.
Calendar	It is used to display calendar and you can add appointments, reminders, and other information in it.
Photos	It is used to store photos and videos and arrange them into albums.
Map	This app helps you to find directions of your destination using GPS (Glabal Positionsing System).
Browser	It helps you to browse the intenet.
Google Play Store	Google store is a place where you can download different apps like Facebook, WhatsApp etc.
Clock	This shows the time and also sets alarm, timers, etc.
Messages	This app helps you to send and receive messages from other people.
Music	This app is used to listen music in mobile phone.
Apple Store	Apple devices have an Apple store where you can download and use various apps.

Difference between a smartphone and a tablet are

Smartphone	Tablet
Screen size is smaller	Has a bigger screen
Watching a movie or making changes to a picture is difficult	Watching videos and making changes to a photo is better and easier due to bigger screen size
You need to charge phones more often	Usually have a longer battery life

Session 3: ICT Tools: Smartphones and Tablets – II

What is mobile layout?

Mobile layout means a visual structure of the mobile which helps the user to interact with the mobile app. To use smartphone, you have to know about basic layout –

- **Power button** – The purpose of power button is to start and shutdown the mobile device.
- **Screen** – Screen helps to perform task by touching with fingers.
- **Back button** – Back button is used to go back to the previous screen in an application.
- **Microphone** – Microphone is used to talk to someone else.
- **Menu button** – This is used to show the options available in a particular app.
- **Home button** – This brings you back to the home Screen.
- **Earpiece** – This helps you to listen to voice calls.

- **Volume button** – This helps you to increase or reduce the sound of your mobile.

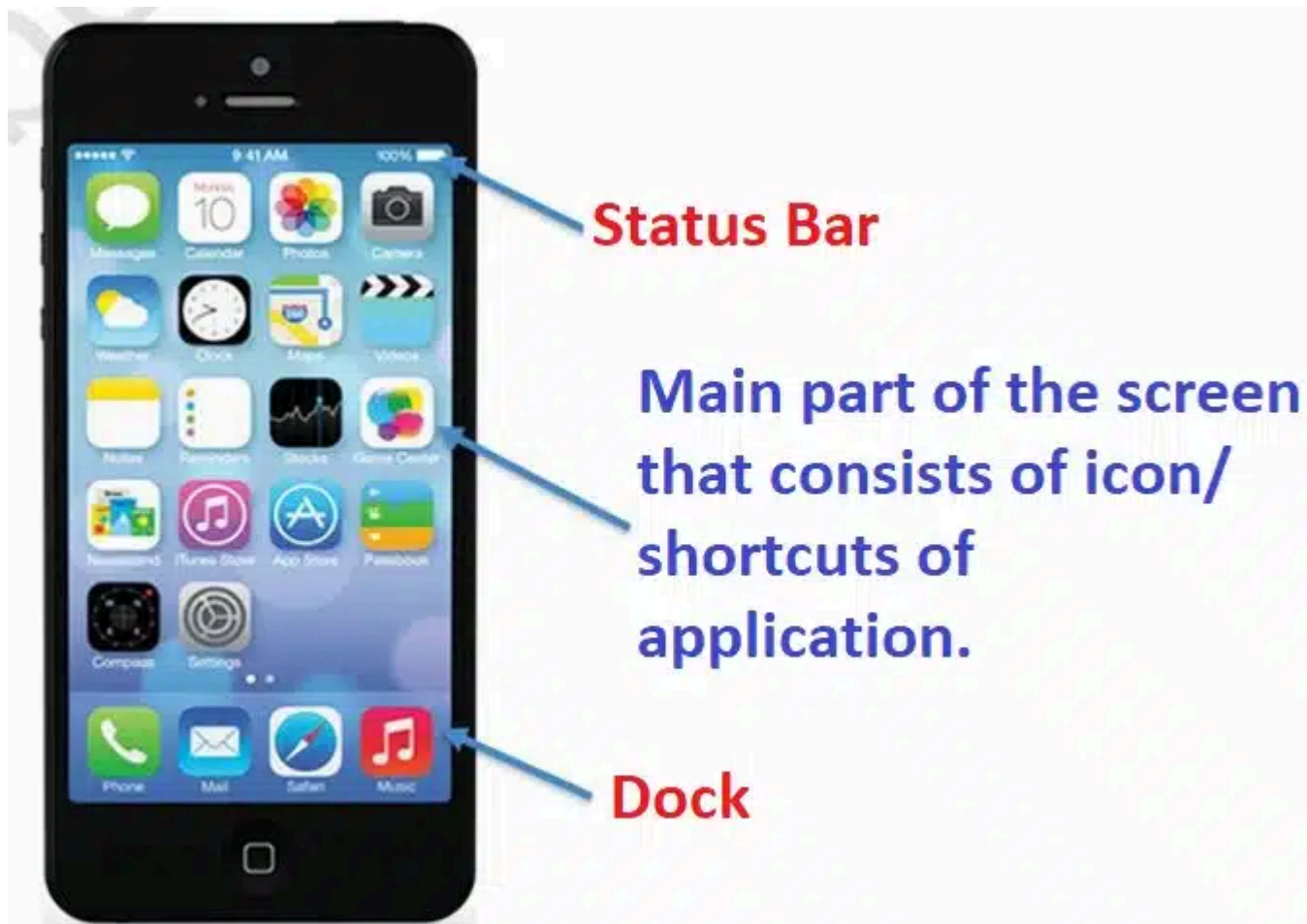
Basic features of a mobile device

The basic features of mobile device are –

	Bluetooth: This is a short-range wireless technology which helps you connect with other devices that are within 30 feet of where you are. Once connected, you can send messages and songs.
	Chargeable Battery: This is a portable power pack which can be recharged. It allows you to use the device anywhere.
	Wi-Fi: This is a wireless network technology that helps you connect to the local area network. This is used to connect to the Internet and work with e-mails, social media and anything to do which requires internet.
	Touch Screen: The output display of a mobile device is a touch screen. This helps you to run apps and type anything into the mobile by simply touching different areas of the screen with your finger.
	Camera: Smartphones and Tablet have a front and back camera to capture pictures and videos.
	Clock: Every mobile device has a clock which can be set according to the time in your country.
	Cellular network connectivity: This provides the network through which you can make calls.
	Global Positioning System: It is a navigation (direction finding) system that helps you to navigate, find direction and maps to specific locations.

Home screen of a mobile device

The home screen is the main screen of a mobile device, there are basically three main parts of the home screen.



- **Status Bar** – The top of the screen displays icons to show network signal, battery life, Wi-Fi and Bluetooth connectivity, 3G/4G or 5G status etc.
- **Main Screen** – The main part of the screen contains icons/ shortcuts of application, this application can place any were.
- **Dock** – This a lower part of the screen which is used to move to a different page in mobile phone.

Basic gestures used

You can interact with your mobile device's touchscreen by making gestures. For example, you can tap the touchscreen to launch an application rather of using a mouse click.

- Tap
- Tap and hold
- Drag
- Swipe
- Double tap
- Pinch

Session 4: Parts of a Computer and peripherals

A computer system is a programmable device made to store and retrieve data as well as carry out logical and mathematical operations to generate useful output. The input unit, central processing unit (CPU), and output unit are its three primary components.

- **Keyboard** – Keyboard is the input unit.

- **Central Processing Unit** – CPU is the processing unit.
- **Monitor** – Monitor is the output unit.

Central Processing Unit

The Central Processing Unit (CPU) is also known as brain of computer, CPU process all the mathematical and logical problems received from the input device and gives the output to the monitor just like a human brain.

There are 3 main parts of the CPU –

- **Control unit** – It controls different parts of computer.
- **Processing unit** – Performs all the mathematical and logical operation.
- **Memory unit** – It work as a storage room, where data is stored temporarily in RAM as well as for a long time (ROM).

Understanding RAM and ROM

RAM stands for Random Access Memory; it is also known as volatile memory. RAM is used to store the data that is currently open in the computer. ROM stands for read-only memory, which stores the data and instruction permanently.

Difference between RAM and ROM –

RAM	ROM
Temporarily stores information	Permanently stores information
Information is lost when computer is switched off	Information is stored when computer is off
Information can be changed easily.	Information can be changed easily.
It is also known as volatile memory	It is also known as non-volatile memory

Motherboard

The motherboard, is also known as the system board, is a main circuit board inside the computer, it connects input, processing, and output devices.

Storage devices

Storage device is used to store the files and folders or digital information which you can carry easily from one place to another place using USB flash drive, hard disk, CD, DVD drives etc.

Computer can measure the amount of data stored on a digital device using bits and bytes. The Bit is a smallest storage unit (Bit represented by 0 or 1). Eight bits combined together and create one byte.

- 8 bits make a 1 byte
- 1024 bytes make a 1 kilobyte (KB)
- 1024 KB make 1 megabyte (MB)
- 1024 megabyte make 1 gigabyte (GB)
- 1024 gigabyte make 1 terabyte (TB)

Ports and connections

Ports are the slots or channels where we insert the cables for the external hard drive, mouse, and keyboard. Ports help us to connect input, output, and storage devices within a computer system. There are several types of ports that you can use to connect an external device.

- **USB Port** – USB stands for Universal serial bus, USB helps to connect peripheral devices, such as a mouse or a keyboard or a printer to a computer.
- **Display port** – It helps to connect monitor, or any display unit to the computer. There are two types of display port Video Graphics Array (VGA) and High-Definition Multimedia Interface (HDMI) depending on the requirement.
- **Audio ports** – It helps to connect microphone, speakers, and headphones to a computer system.
- **Ethernet port** – Ethernet port is used for connecting the system to high-speed Internet cable.
- **Power port** – Power port is used for connecting the computer system to the power supply.

Session 5: Basic Computer Operations

A computer system consists of two main parts—the hardware and the software.

- **Hardware** – The physical parts that we can see, and touch are called the hardware. Example, Keyboard, monitor, CPU etc.
- **Software** – The software that we cannot see makes the hardware work the way we want. for example, Operating System, LibreOffice Writer etc.

Basic functions performed when a computer starts

When we switch on a computer, computer automatically runs a basic program called BIOS (Basic Input/Output System). The BIOS first runs a self-test. If the self-test shows that the system is fine, the BIOS will load the Operating System.

Using the keyboard

A keyboard helps to insert text, number and command into the computer. There are various types of keys available on keyboard –

- **Function keys** – Function keys start from F1 to F12 to perform specific work. for example, F1 key is used to open help program in computer.
- **Control keys** – Control keys perform special functions depend on the purpose. for example, Control (CTRL), Shift, Spacebar and ALT.
- **Enter key** – Enter and Return keys help to move the cursor to the beginning of a line.
- **Punctuation keys** – Colon (:), semicolon (;), single quotation marks (‘ ’), double quotation marks (” “) and question marks (?) are known as punctuation keys.
- **Navigation keys** – The arrow keys, END, HOME, Page Up and Page Down are known as navigation keys.
- **Command keys** – Insert (INS), Delete (DEL) and Backspace are known as command keys.
- **Windows key** – Pressing this key opens the Start menu.

Using a mouse

A mouse is a small device that you can use to move, select, and open items on your computer screen.

- **Roll over/hover** – Moving a mouse over a file or folder in file explorer, it will show the details of file and folder.
- **Point and click** – The mouse allows you to select an item on the screen.
- **Drag and drop** – move the item to a new location.
- **Double-click** – When we double-click the file icon, it will open the file.

Session 6: Performing Basic File Operations

What is files and folders in computer system?

All the information in computer stored in files, every file name has a file name and extension that identifies the file type. The file name and file extension are separated by a period or a 'dot'. for example, notepad file has an extension '.txt', image file has an extension '.jpg', sound file has an extension '.mp3'. **Folder is a collection of files and sub folder.**

Session 7: Communication and Networking – Basics of Internet

What is Internet?

The Internet is a global network of interlinked computers and electronics devices. It is a vast network system that links computers worldwide. Every piece of information on the internet is kept in a webpage on the internet.

Uses of the Internet

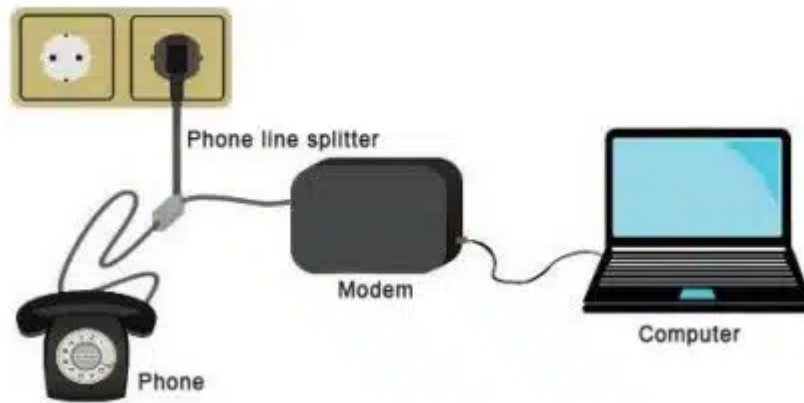
There is no limit to what you can do online, but some of the most common uses of internet are –

- Searching Information
- Online Shopping
- Online Banking
- Online learning
- Entertainment
- E-mail and chatting
- Social networking

Connecting to the Internet

To connect to the Internet, you need:

- A device, such as a computer, laptop or mobile.
- A connection device, i.e., a modem.
- An Internet Service Provider (ISP) – the telephone company which provides the Internet connectivity.



Types of connections

There are two ways for connecting your computer to the internet:

- **Wired connections** – When computer is connected to the internet using wire (Fibre optical, Ethernet cable or Co-axial cable) is called wired connection.
- **Wireless connection** – When the computer is connected without wire (Wi-Fi, Cellular, Satellite communication etc.) is called wireless connection.

What is Bandwidth?

Bandwidth is the maximum amount of data that can be transmitted over a network in a certain amount of time. The bandwidth commonly measured in bits per second or bytes per second. For example, If bandwidth having a capacity of 100 MBPS, it means that network cannot transfer the data more than 100 MBPS.

The data transfer speed can be calculated in terms of bits per second, for example,

- 1 Kbps (kilo bits per second) = 1000 bits per second
- 1 Mbps (mega bits per second) = 1000 kilo bits per second
- 1 Gbps (giga bits per second) = 1,000 mega bits per second

Session 8: Communication and Networking – Internet Browsing

What is World Wide Web?

World wide web is a hypertext based information, files, images and videos stored in a web pages and accessible through the internet. This hypertext helps you to navigate between the web pages.

World Wide Web is made up of –

- A web page
- A web browser
- A system to transfer information between the web browser and the web pages

What is Web page?

A web page is a document that is shared online and stored on a web server. Every web page has a unique address which is known as domain name. For example, the address of CBSE official website is <https://cbse.nic.in>, to open web page you required web browser on your computer.

What is hyperlinks?

Websites contain multiple hyperlinks; these hyperlinks are highlighted and underlined words with a blue color. This hyperlink helps you to navigate from one page to another page or another section of the same page.

What is Internet browser?

An Internet Browser is an application or a software program on your computer or laptop which helps you visit the various websites. Some examples of **browsers are Google Chrome, Mozilla Firefox and Microsoft Edge.**

Important parts of a web browser –

- Address bar (Alt+d)
- Tabbed browsing (Ctrl+T)
- Back and forward buttons (Alt+Left or right arrow keys)
- Refresh button (F5)

Session 9: Communication and Networking – Introduction to E-Mail

What is e-mail?

Electronic mail, or e-mail, is a quick way of sending messages to people using the Internet. An e-mail address is made up of two parts separated by the @ symbol. The general format of an e-mail address is local_part@domain. for example, “anuraganand2017@gmail.com”, the part before the @ is created by the user, and the part after the @ is the domain name of the e-mail service provider.

Difference between postal mail and e-mail

Postal Mail	e-Mail
Postal mail address is a geographical location.	e-mail address is a digital location
Postal mail is written with paper and pen.	An e-mail is written on a computer or mobile device.
Postal mail is sent by bus, air, train.	e-mail travels in a digital format over an Internet connection.
Postal mail is brought by the postman and dropped into the letterbox	E-mail comes into the Inbox of the receiver's e-mail account.

Advantages of e-mail

- e-mails can be sent to multiple users along with the attachments.
- e-Mail are fast as they reach anyone around the world immediately
- e-Mail most services are free
- e-Mail are environment friendly, as no paper is used.

Session 10: Communication and Networking – Creating an e-Mail account

To set up an e-mail account, there are a number of popular providers to consider –

- **Gmail** (run by Google)
- **Outlook mail** (run by Microsoft)

- **Yahoo mail** (run by Yahoo)

Steps to open an e-mail account on Gmail

- **Step 1:** Open <https://gmail.com>
- **Step 2:** Click on “Create an account”
- **Step 3:** Fill the basic information, including username and password
- **Step 4:** Click on Next button
- **Step 5:** Google “Welcome page will appear”
- **Step 6:** Click on “Continue to Gmail”

Session 11: Communication and Networking – Writing an e-Mail

Writing an e-Mail

Gmail is a free e-mail application follow the following steps to writing an e-Mail.

- **Step 1:** Connect to the Internet
- **Step 2:** Open Gmail “<https://gmail.com>” in google chrome
- **Step 3:** Type e-Mail ID and password
- **Step 4:** Click on “Sign in” button
- **Step 5:** Click on “COMPOSE” button on the gmail home page.
- **Step 6:** Write the information including Subject
- **Step 7:** Write e-Mail address
- **Step 8:** Click on “Send” button

Managing Folders in e-Mail Home page

There are different folders that could help you manage and organise your e-Mails.

- **Inbox** – The Inbox holds e-mails.
- **Sent** – The Sent folder holds e-mails that you have sent to people
- **Drafts** – The Drafts folder holds all the e-mail you have started writing, but not yet sent
- **Junk/Spam** – The Junk/Spam folder holds e-mails that may not be useful
- **Trash** – The Trash can or folder holds e-mails that you no longer need and have deleted.